

Constructing and Solving Equations — Paper A

MEDIUM • NON-CALCULATOR • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every equation on this paper has a whole-number solution unless it says otherwise.
- Whatever you do to one side, do to the other. Write the operation beside each line, like “ -4 ” or “ $\div 3$ ”.
- Strip the numbers off the x first and divide last. Dividing early makes you divide every term.
- Every answer can be checked by putting it back into the original equation. Do it every time — it costs ten seconds and it catches nearly every error.
- The mark for each question is shown in square brackets on the right.

SECTION 1 – ONE STEP, THEN TWO

- 1.** Complete the table. Each empty box is worth one mark. In the last column, show the substitution that proves your answer is right. [6]

Equation	Solution	Check (substitute back)
$x + 9 = 17$		
$5x = 45$		
$3x + 2 = 20$		

The check column is not decoration — it is where you find out whether you were right before the marker does.

- 2.** Solve:

(a) $x - 6 = 11$ [1]

(b) $7x = 56$ [1]

(c) $x \div 4 = 6$ [1]

(d) $x + 13 = 5$

[1]

3. Solve:

(a) $2x + 5 = 17$

[1]

(b) $4x - 3 = 25$

[1]

(c) $5x + 8 = 8$

[1]

(d) $3x - 11 = 4$

[1]

SECTION 2 – BOTH SIDES, AND BRACKETS

4. Solve, showing each step:

(a) $5x = 2x + 18$

[2]

(b) $7x - 4 = 3x + 12$

[2]

5. Solve:

(a) $2(x + 5) = 16$

[2]

(b) $3(x - 4) = 21$

[2]

SECTION 3 – BUILDING THE EQUATION YOURSELF

6. I think of a number, multiply it by 5, then subtract 3. The result is 32.

(a) Using n for the number, form an equation.

[1]

(b) Solve it.

[1]

(c) Check your answer against the original instructions.

[1]

7. A rectangle has width x cm and length $(x + 3)$ cm. Its perimeter is 26 cm.

(a) Form an equation in x .

[1]

(b) Solve it.

[1]

(c) Write down the width and the length, and check the perimeter.

[1]

8. Ali has n marbles. Ben has 7 more than Ali. Together they have 41 marbles.

(a) Form an equation in n .

[1]

(b) Solve it, and state how many marbles each boy has.

[1]

End of paper. Check every answer has a unit or a form the question actually asked for.